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MILWAUKEE VIETNAM
Traceforward

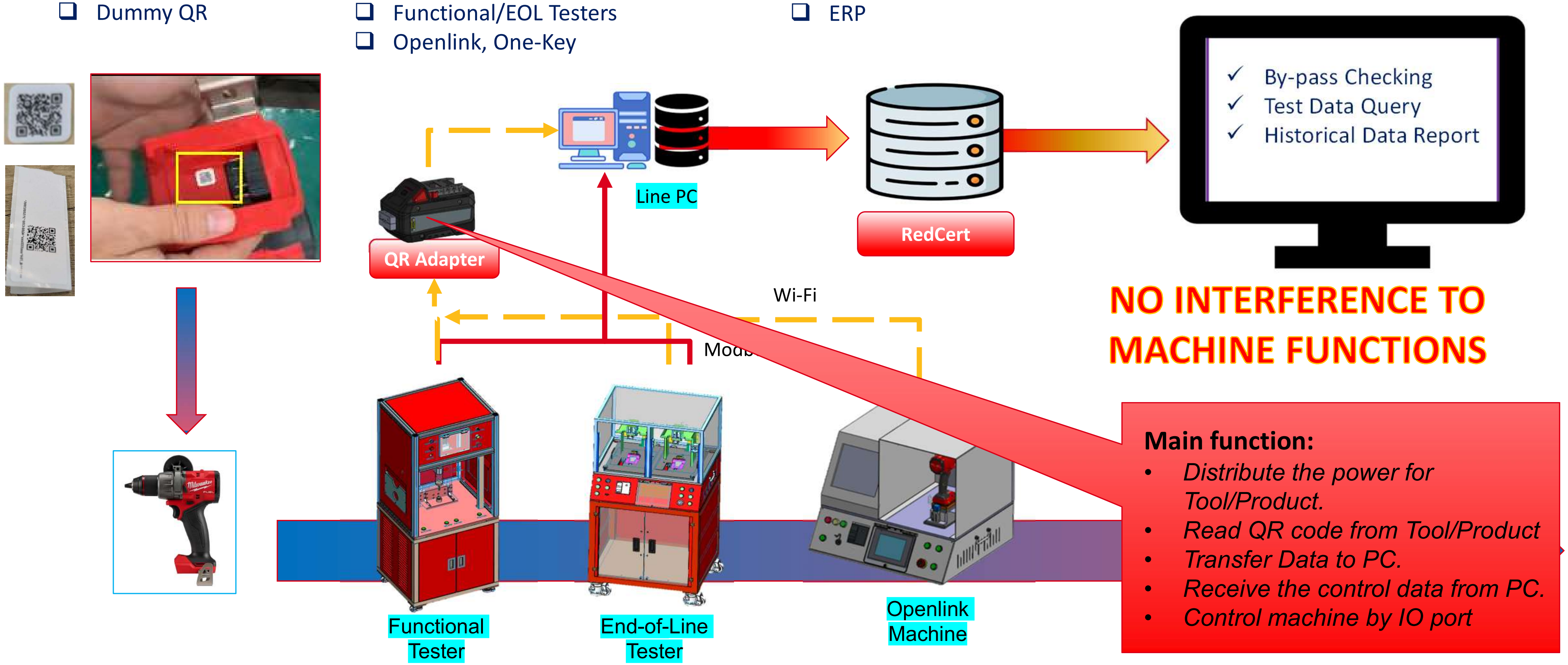
VN - QR Adapter

Milwaukee Trace Forward

Traceforward

What is QR Adapter?

Identification	Connectivity	Centralize Data	Traceability
❑ Tool-ID ❑ Dummy QR	❑ QR Label Reader Adapter ❑ Functional/EOL Testers ❑ Openlink, One-Key	❑ RedCert Database ❑ ERP	



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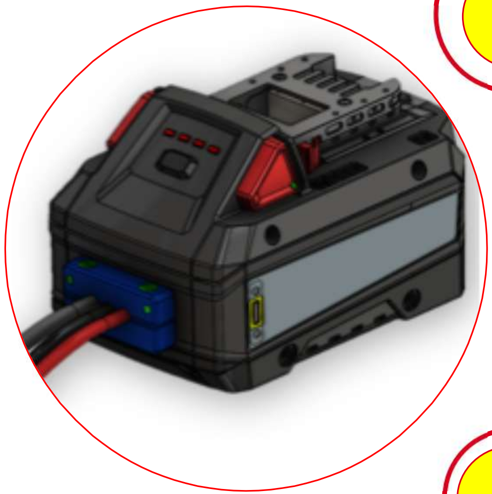
Why is QR Adapter?



1 Automatic QR reading	No extra machine cycle time
2 Multiple function	QR scanning, power conversion, Machine interaction
3 Flexibility	Customizable size and distance, with remote server connectivity
4 Low cost	Using components and self-assembly with internal resources

VN - QR Adapter

Initial requirement

1. Read QR code
2. Send QR code to PC
3. Power converter for tool
4. Easy for change over & Replacement
5. Dual channel to communication with both Trace PC and Openlink PC
6. Control Openlink machine via IO port
7. Get signal from Openlink machine for result
8. Standard to be deploy for single process without any PC
9. Low cost
10. Have ability to integrated function to read MPBID without middleware

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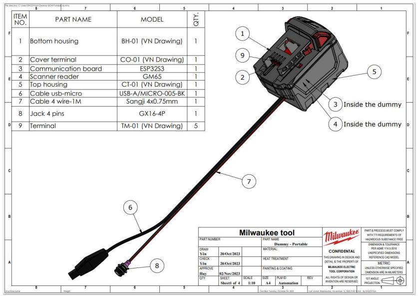
Development Stages

Design	Prototype	Pilot Run	Apply 1st MP	Standardization	Scale up	CI/CD
Q2 2023	Q1 2024	Q2 2024	Q3 2024	Q4 2024	Q1 2025	Now



ITEM NO.	PART NAME	MODEL	QTY.
1	Bottom housing	BH-01 (VN Drawing)	1
2	Cover terminal	CO-01 (VN Drawing)	1
3	Communication board	ESP32S3	1
4	Scanner reader	GM65	1
5	Top housing	CT-01 (VN Drawing)	1
6	Cable usb-micro	USB-A/MICRO-005-BK	1
7	Cable 4 wire-1M	Sangji 4x0.75mm	1
8	Jack 4 pins	GX16-4P	1
9	Terminal	TM-01 (VN Drawing)	5

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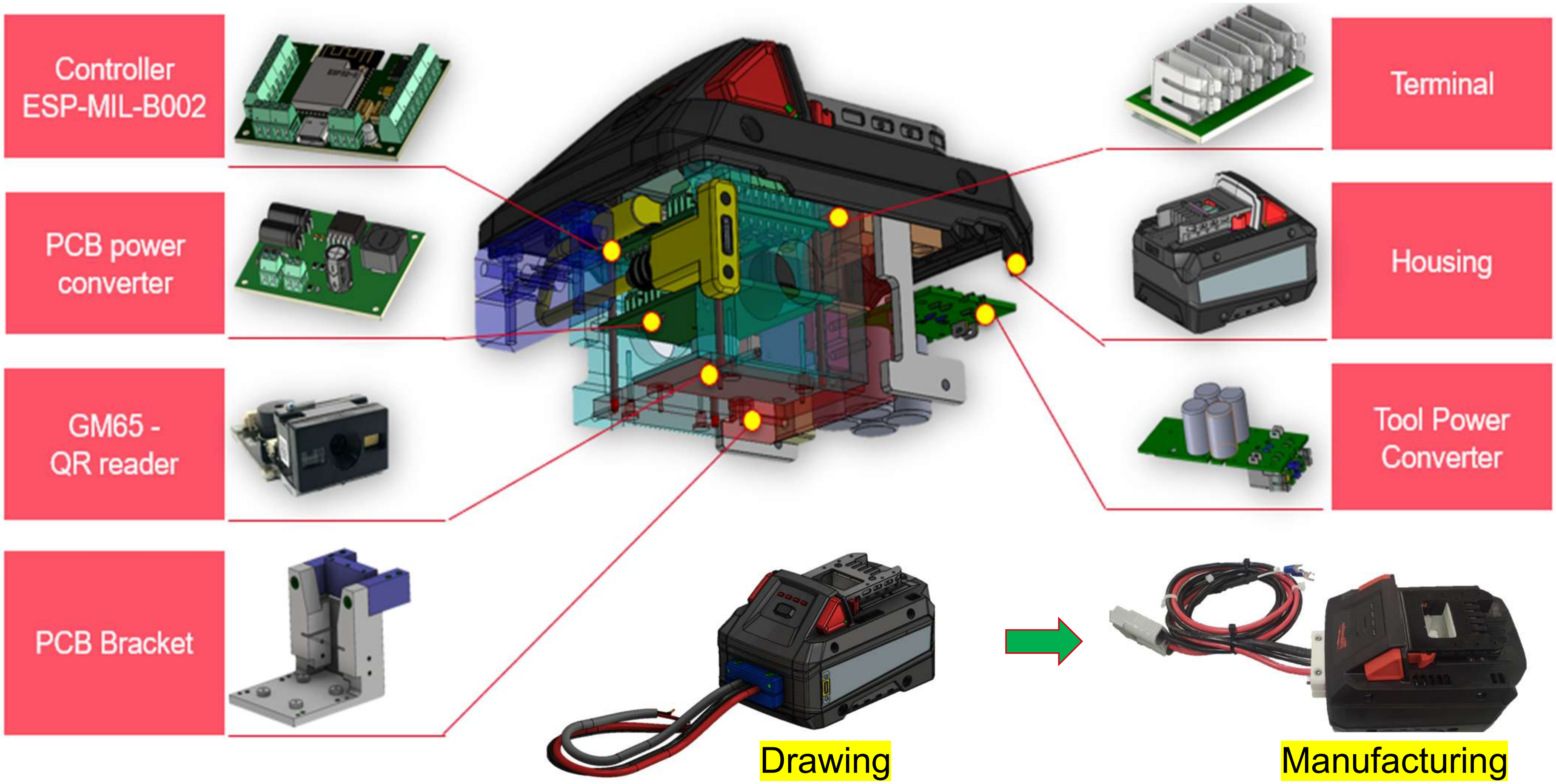
Hardware Explored View



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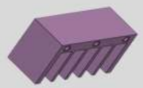
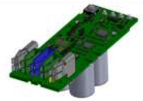
Part list

No.	Name	Spec	Description	Brand	Unit	Note	Picture
1	Scanner reader	GM65 2D/3D Barcode scanner	Read QR from tool	Hangzhou Grow Technology	1	Final	
2	Control board	ESP32-VNMIL-01	+ Receive QR code from GM65. + Connect wifi and send/Receive data to/from PC + Controller (Input, output)	TTI-Dien tu tuong lai	1	Final	
3	Top housing	319309002	TOP HOUSING SUB ASSEMBLY FOR M18 HD LEAP DESIGN **(130433001DG9C-02_M18 G9 FORGE XC 8.0AH PACK WITH AMPACE CELL)**	MIL	1	Final	
4	Bottom housing	546863002	LEAP DESIGN BOTTOM HOUSING PLASTIC INJECT WITH OVERMOLD **(130433001DG9C-02_M18 G9 FORGE XC 8.0AH PACK WITH AMPACE CELL)**	MIL	1	Final	
5	Electric wire clip	Drawing	Jig	Manufacturing	1	Final	
6	Reader hanger part	Drawing	PCB board hanger	Manufacturing	1	Final	
7	Left plate	Drawing	Board jig	Manufacturing	1	Final	
8	Right plate	Drawing	Board jig	Manufacturing	1	Final	
9	Terminal Pin	634553014-TTI	Tool connector	MIL	5	Final	
10	PCB for terminal	Drawing	Terminal Hanger	Manufacturing	1	Final	
11	Insulating barrier	Drawing	Insulate the terminal pins	Manufacturing	1	Final	
12	M18 Milwaukee Dummy pack PCBA	280353005	Power Converter and distribute for Tool	MIL	1	Final	
13	Fixing adapter	Drawing	Adapter jig	Manufacturing	2	Final	

VN - QR Adapter
ESP32 - Software Module

Name: **ESP-MIL-B002**
Core CPU: **ESP-32S, ESP32-C5, ESP32E**
Programe language: **C++**
IDE: **Arduino/ VS code**

1 Local Data Management
- Storage WiFi infor, Host info, operation mode, Adapter type, Assetcode
- Update the Operation Info

2 Wi-Fi Management
- Connect and reconnect the wifi network

3 Socket TCP management
- Connect and reconnect the host PC
- Send message from QR scanner (GM65) and openlink App to Host.
- Receive and process the data from Host

4 QR reader management
- Receive the QR reader via serial port
- Extract and encrypt the data before send to host via Socket

5 Serial connection management
- Connect to Openlink PC/App
- Send data from host socket to openlink App
- Receive the data from Openlink app

6 Pokayoke Logical
- Logic for validate QR code
- Logic for Trace added

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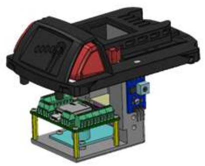
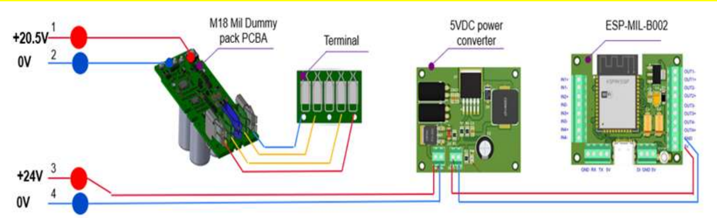
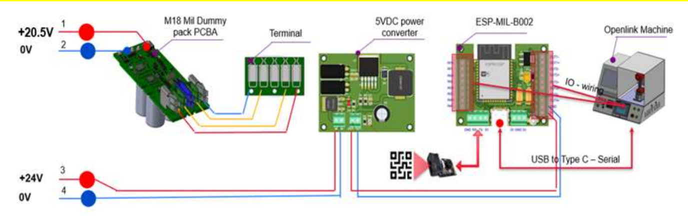
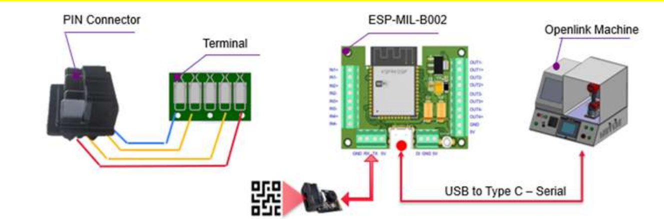
Current state of deployment

EOL machine		Openlink machine	
Portable	Customize	Customize	Portable
			
1. Read QR code and send to Trace PC. **→ Can be use Wire or Wireless**		1. Read QR code and Send to Trace PC. 2. Delivery QR code and product result from Trace system to Onekey system in Realtime. 3. Delivery Job number, Line number, MPBID information from Openlink to Trace PC. 4. Delivery command poka-yoke from Trace PC to PLC via IO port. **→ Need 3 communication channel.**	1. Read QR code and Send to Trace PC. 2. Delivery QR code and product result from Trace system to Onekey system in Realtime. 3. Delivery Job number, Line number, MPBID information from Openlink to Trace PC. **→ Need 2 communication channel.**







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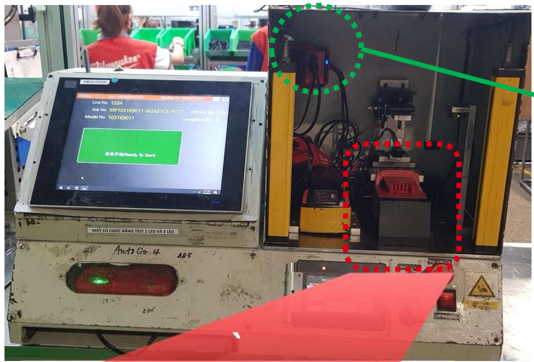
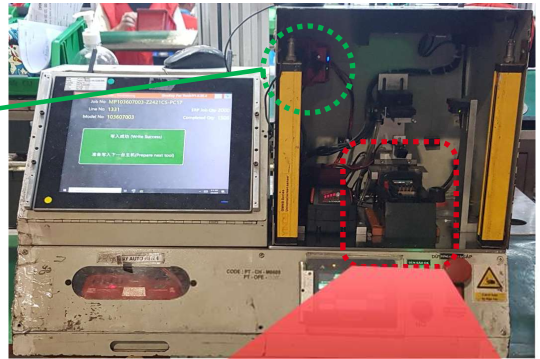
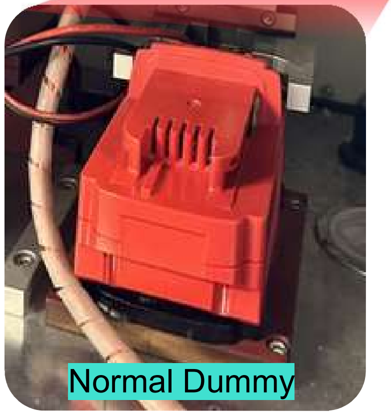
Openlink machine – hardware changed



2800-20 MPBID adapter
(Don't change MPBID adapter)





Before

After

Auto write Openlink Machine Modification

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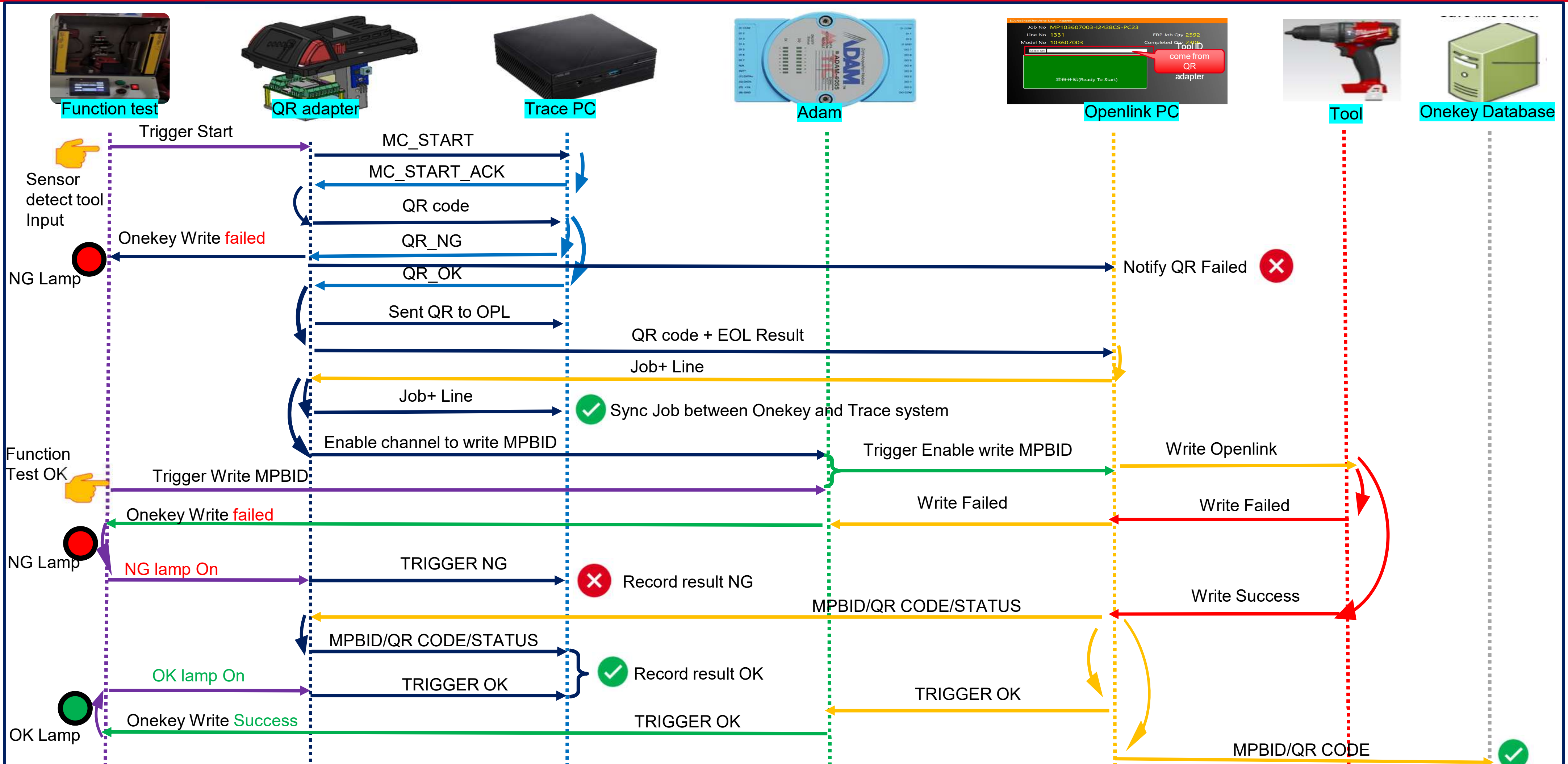
Openlink machine Auto – Machine/PC/QR adapter data flow



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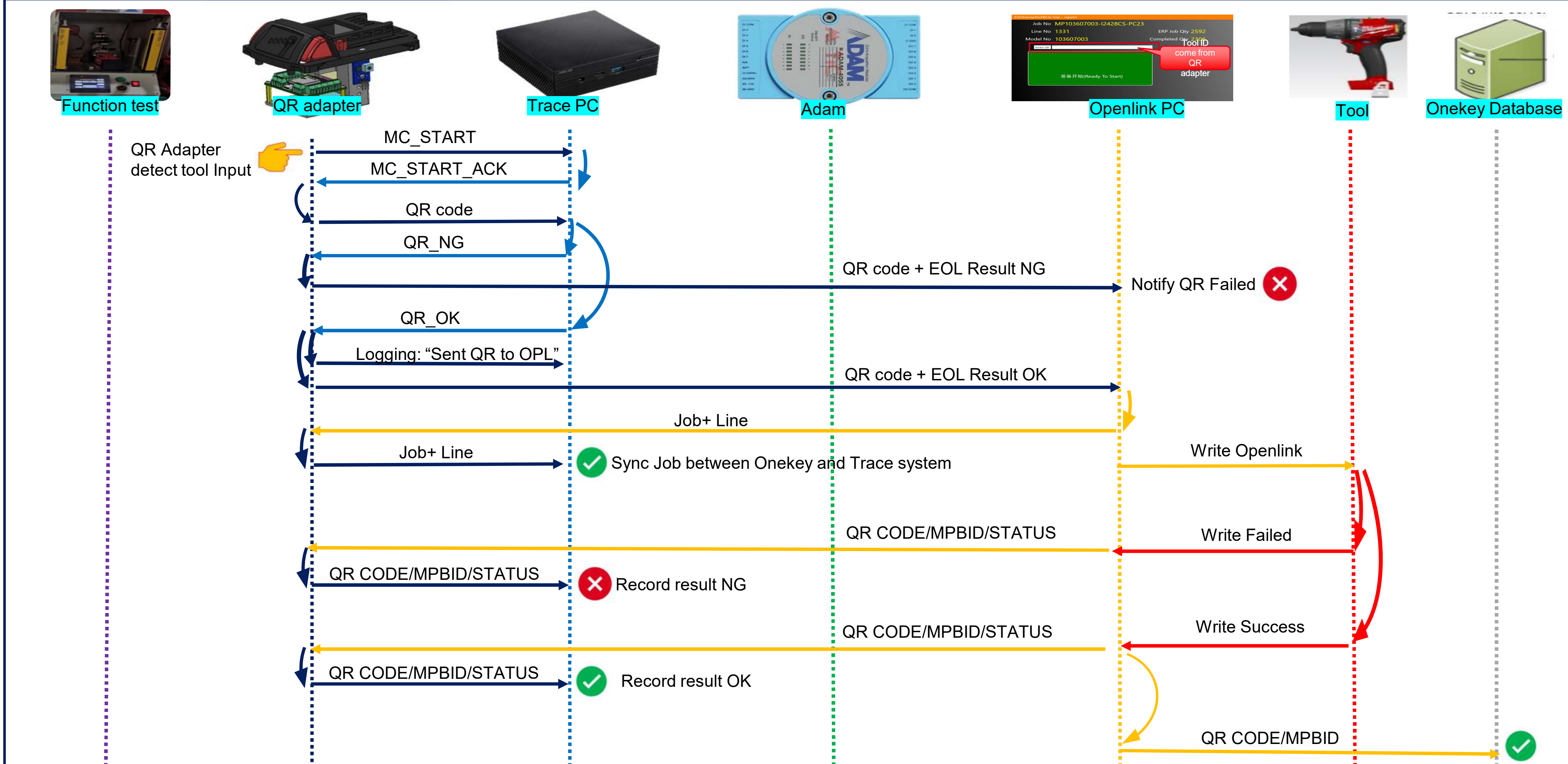
Openlink machine Manual – Machine/PC/QR adapter data flow



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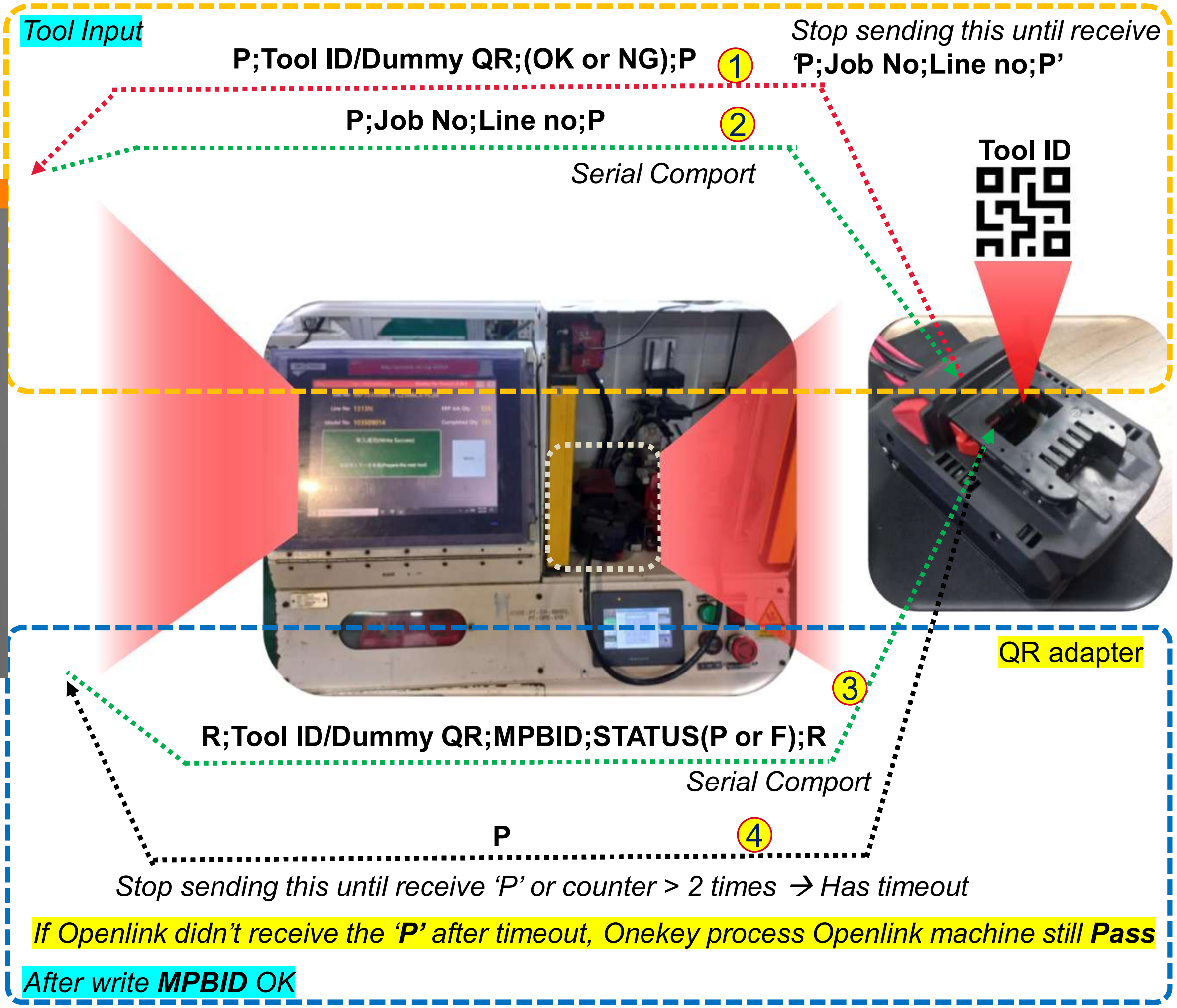
Openlink machine – OL App and QR adapter interraction



Display Result of the Tool ID/Dummy ID in Openlink App:

OK: Pass Previous station

NG: Failed Previous station

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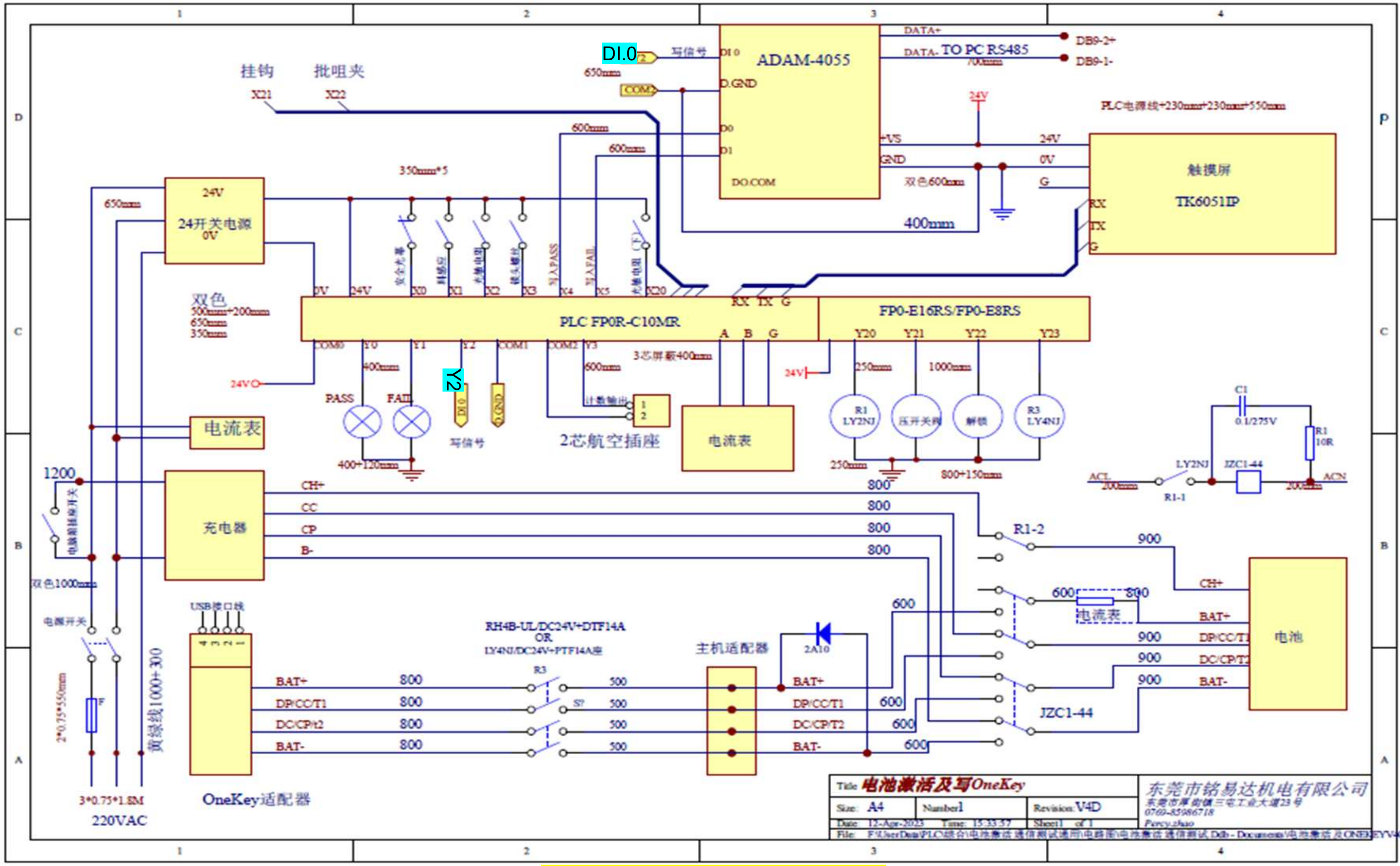
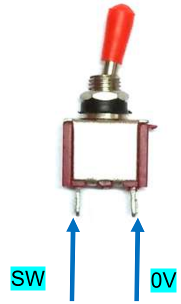
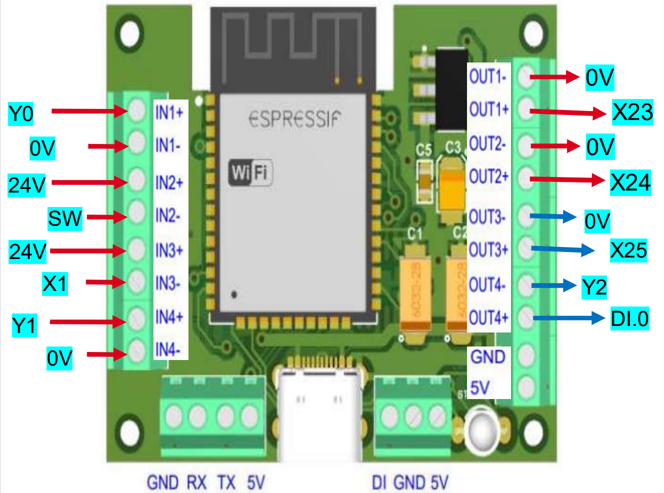
Openlink machine – Electrical modification







OpenLink EE Schematic

ESP-MIL-B002

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THE END

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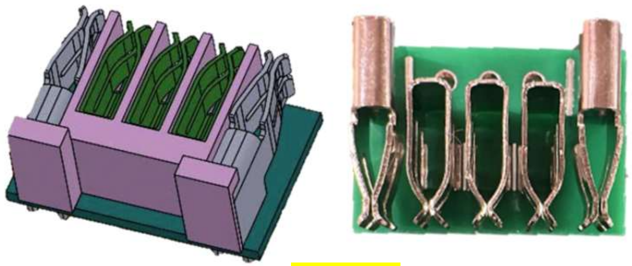
Adapter upgrade proposal

As is



To be

Openlink App start → Send Job + Line → Sync back to EOL machine

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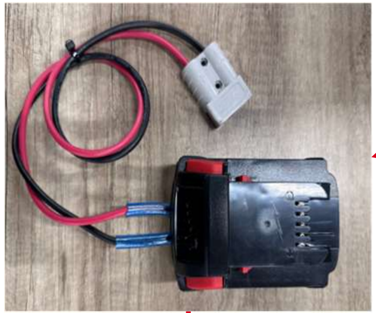
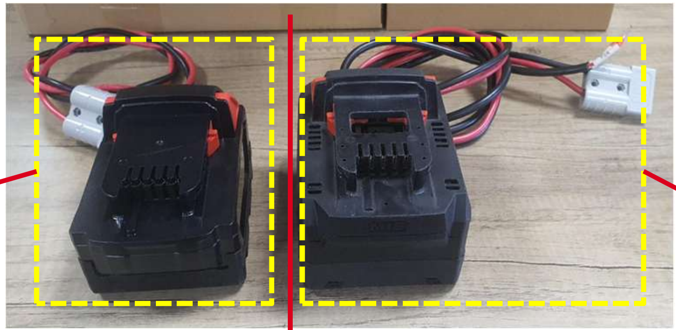
Adapter upgrade proposal





Normal Dummy
- DUMMY_FIRMWARE_VERSION
- DUMMY_IDENTIFIER

QR Adapter
- DUMMY_FIRMWARE_VERSION
- DUMMY_IDENTIFIER
- QR_ADAPTER_MAC_ADDRESS
- QR_ADAPTER_FIRMWARE_VERSION

Normal Dummy

QR Adapter